

# Juwon Kim

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## Research Interests

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Kernel, Filesystem, Flash storage, Device firmware (Flash Translation Layers), and Garbage collection.

## Education

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**Ph.D. in Electrical Engineering** Mar. 2023 – Aug. 2026  
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Advisor: Prof. Youjip Won

**M.S. in Electrical Engineering** Mar. 2021 – Feb. 2023  
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Thesis: *IPLFS: Log-Structured Filesystem without Garbage Collection*
- Advisor: Prof. Youjip Won

**B.S. in Electrical Engineering** Mar. 2016 – Feb. 2021  
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Individually Designed Major in Computer Science and Physics

## Publications

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- [1] **[ACM SOSP '26]** Juwon Kim, Hwanseok Yoo, Seung Won Yoo, and Youjip Won, “GAZE: Rearchitecting Log-Structured Filesystem for Garbage Collection on Zoned Namespace Flash Storage”, accepted for publication in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)* (acceptance rate: 15%), Prague, Czechia, Sep.–Oct. 2026.
- [2] **[IEEE NVMSA '26]** Jihoon Seok, Juwon Kim, Youjip Won, “LeaST: Learned Sector Translation for Skipped Physical Page Address”, in *Proc. of IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA)*, Aug. 2026.
- [3] **[USENIX FAST '25]** Juwon Kim, Seungjae Lee, Joontaek Oh, Dongkun Shin, Youjip Won, “D2FS: Device-Driven Filesystem Garbage Collection”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [4] **[USENIX FAST '25]** Jieun Kim, Joontaek Oh, Juwon Kim, Seung Won Yoo, Youjip Won, “OPIMQ: Order Preserving IO Stack for Multi-Queue Block Device”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [5] **[ACM SAC '25]** Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, Joontaek Oh, Youjip Won, “lwFSCK: Light-weight Filesystem Check”, in *Proc. of ACM Symposium on Applied Computing (SAC)*, Catania, Italy, Mar. 2025.
- [6] **[USENIX ATC '22]** Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection”, in *Proc. of USENIX Annual Technical Conference (ATC)* (acceptance rate: 16%), San Diego, USA, Jul. 2022.

## Posters and Workshops

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- [1] **[ACM HotStorage '26]** Hwanseok Yoo, Juwon Kim, Youjip Won, “Manycore Scalable Logging for Log-Structured Filesystem,” in *Proc. of ACM Workshop on Hot Topics in Storage and*

*File Systems (HotStorage)*, Prague, Czechia, Sep. 2026.

- [2] **[ACM SOSP '25] Juwon Kim**, Hwanseok Yoo, Youjip Won, “Accelerating Filesystem Garbage Collection for Zoned Flash Storage,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [3] **[ACM SOSP '25] Hwanseok Yoo, Juwon Kim**, Youjip Won, “Manycore Scalability of Modern Log-Structured Filesystem: Revisited,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [4] **[USENIX FAST '22] Juwon Kim**, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection,” in *Proc. of USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2022 (**Poster**).

## Patents

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- [1] Youjip Won and **Juwon Kim**, “Method and Device for Managing Host Area,” Korean Patent Application No. 10-2025-0174419, filed Nov. 2025.
- [2] **Juwon Kim** and Youjip Won, “Storage Device Performing Garbage Collection,” Korean Patent Application No. 10-2025-0090393, filed Jul. 2025. (*Joint filing: SK Hynix & KAIST*)

## Awards & Honors

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| • Student Scholarship, ACM SOSP, Czechia                             | Sep. 2026 |
| • Student Grant (Full), USENIX FAST, USA                             | Feb. 2025 |
| • NVRAMOS 2023 OpenSSD Best Practice Award, Gold Prize, KIISE, Korea | Nov. 2023 |
| • Student Grant (Full), USENIX FAST, USA                             | Feb. 2022 |

## Research Experience

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| <b>Postdoctoral Researcher</b><br>Operating Systems Laboratory, KAIST | <i>Sep. 2026 – Present</i><br>Daejeon, Korea |
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| <b>Visiting Researcher</b><br>SK Hynix | <i>Jul. 2025 – Aug. 2025</i><br>Bundang, Korea |
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### Filesystem Garbage Collection for Zoned Mobile Storage

- Investigated filesystem garbage collection overhead on mobile platforms and explored host-side optimizations for zoned UFS devices.

*Zoned UFS, F2FS, Garbage collection, Mobile storage, Linux kernel*

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|----------------------------------------------------------------------------------|------------------------------------------------|
| <b>Graduate Researcher (Ph.D. + M.S.)</b><br>Operating Systems Laboratory, KAIST | <i>Mar. 2021 – Aug. 2026</i><br>Daejeon, Korea |
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### Log-Structured Filesystem for Large-Zone Reclamation

- Rearchitected a log-structured filesystem for efficient reclamation of large zones in Zoned Namespace flash storage, eliminating fundamental CPU and I/O bottlenecks in conventional filesystem garbage collection.

*F2FS, Zoned Namespace SSD, Garbage collection, Linux kernel, Memory management*

### Device-Driven Filesystem Garbage Collection

- Designed a host–device cooperative filesystem architecture that delegates filesystem garbage collection to the storage device, reducing host CPU utilization and host–device data movement.

*Filesystem, Flash translation layer, Garbage collection, Host–device cooperation, NVMe*

### Log-Structured Filesystem without Garbage Collection

- Designed a log-structured filesystem that eliminates filesystem-level garbage collection by virtually expanding the filesystem partition over additional physical storage space.

*F2FS, Log-structured filesystem, Garbage collection, Flash storage, Linux kernel*

**Undergraduate Research Intern**  
Operating Systems Laboratory, KAIST

*Jan. 2020 – Feb. 2021*  
Daejeon, Korea

#### **Log-Structured Filesystem Design**

- Investigated log-structured filesystem design and filesystem garbage collection mechanisms under Prof. Youjip Won.

*Filesystem, F2FS, Linux kernel, Flash storage*

### **Work Experience**

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**Machine Learning Engineer Intern**  
Penta Security Systems Inc.

*Jun. 2018 – Jan. 2019*  
Seoul, Korea

#### **CAN Data-Based Anomaly Detection for Electric Vehicles**

- Designed and implemented a GRU-D autoencoder for detecting anomalous vehicle behavior from CAN bus data.

*Machine learning, GRU-D, Autoencoder, CAN bus, Anomaly detection ““*

### **Research Projects**

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**Ultra-Low-Latency Storage-Driven I/O Subsystem for Large Language Models** *Jul. 2026 – Present*

InnoCORE, Ministry of Science and ICT, Korea

**Order Preserving Distributed Storage System**  
NRF, Korea

*Jun. 2025 – Present*

**Optimizing Filesystem Garbage Collection for Zoned Flash Storage** *Sep. 2024 – Present*  
SNU-SKH Solution Research Center, SK Hynix, Korea

**Device-Driven Filesystem Garbage Collection**  
SNU-SKH Solution Research Center, SK Hynix, Korea

*Sep. 2023 – Aug. 2024*

**Collection and Integrated Analysis of Automotive Inter/Intra System Artifacts** *Jan. 2024 – Feb. 2025*  
IITP, Korea

**Future Scalable OS: Log-Structured Filesystem without Garbage Collection** *Jan. 2021 – May 2023*  
IITP, Korea

### **Teaching Experience**

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- Head TA, Operating Systems and System Programming for EE (EE415), KAIST *Sep.–Dec. 2025*
- TA, Special Topics in Computer Engineering (EE817), KAIST *Feb.–Jun. 2025*
- TA, Operating Systems and System Programming for EE (EE415), KAIST *Sep.–Dec. 2024*
- Head TA, Programming Structure for EE (EE209), KAIST *Feb.–Jun. 2024*
- TA, Programming Structure for EE (EE209), KAIST *Sep.–Dec. 2023*
- TA, Programming Structure for EE (EE209), KAIST *Feb.–Jun. 2023*

### **Service**

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- Artifact Evaluation Committee, USENIX ATC '26 *Sep. 2026 – Oct. 2026*

- Artifact Evaluation Committee, USENIX FAST '27 Spring Cycle Jun. 2026 – Jul. 2026
- Lab Manager, Operating Systems Laboratory, KAIST Jan. 2026 – Jul. 2026
- Program Chair Assistant, USENIX FAST '24 Sep. 2023 – Sep. 2024

## Technical Skills

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**Systems:** F2FS, flash translation layer, SSD firmware, NVMe driver, block I/O layer, ZNS, Linux page management subsystem

**Languages:** C, Python, C++, SQL, Verilog

**Tools:** git, vim, gnuplot, ctags, cscope, ftrace, blktrace